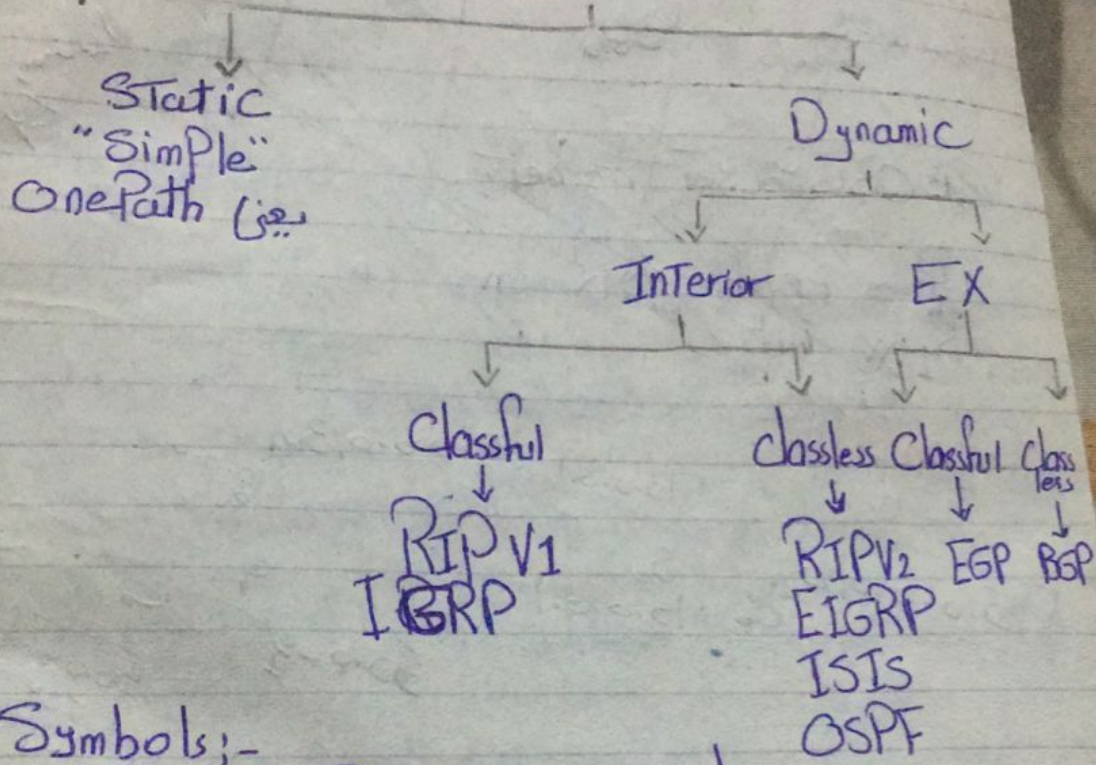


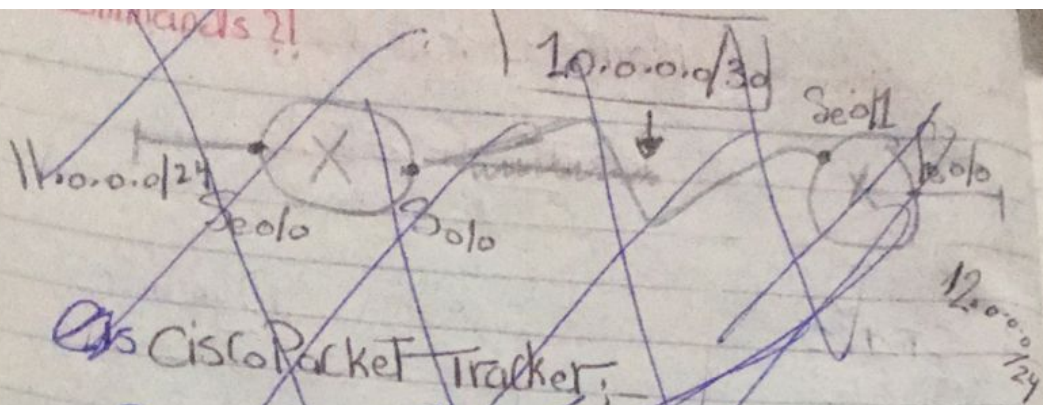
Sec 1 Router Protocols

Routing Protocols



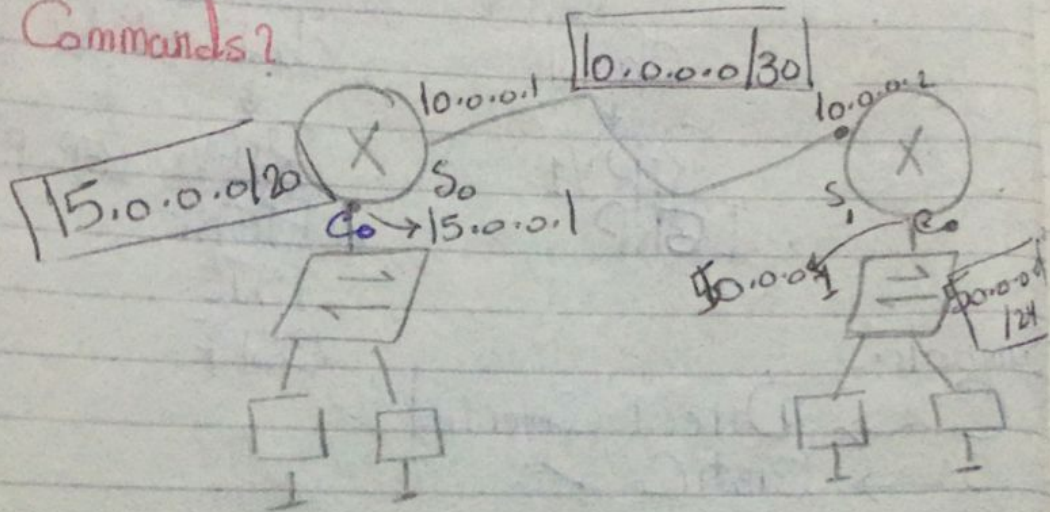
Symbols:-

C ~> Direct Connected ✓
S -> Static ✓



~~as Cisco Packet Tracker;~~
~~R1 > enable~~
~~R1 # int~~

Commands?



Cisco Packet Tracer:-

R1 > enable (en)

R1 # Config t

R1 (Config) # int eth0

R1 (Config-if) # IP address 15.0.0.1 255.255.255.0

R1 (Config-if) # no shutdown → Exit

R1 (Config) # int S0

R1 (Config-if) # IP address 10.0.0.1 255.255.255.0

R1 (Config-if) # no shutdown → Exit

Same on S1 10.0.0.2

Same on E0 50.0.0.1

R1 (Config) # IP route 15.0.0.0
255.255.255.0 S1

R1(Config) # ip route 50.0.0.0 255.255.255.0 S0

R1(Config) # Ctrl Z

R1 # sh ip route ✓✓

ENSP

<Huawei> Sys

[Huawei] int eth0

[Huawei-int] ip address 15.0.0.1 24

["] quit

[Huawei] int S0

[Huawei-int] ip address 10.0.0.1 30

Same on S1 10.0.0.2

Same on e0 50.0.0.1

[Huawei] ip route-static 50.0.0.0 S0

Same on S1

[Huawei] display ip routing-table

Ans

RIP

• RIP

• Adm

• Per

• Ma

9/15/23

SECTION 2

RIP

RIP → Routing Information Protocol.

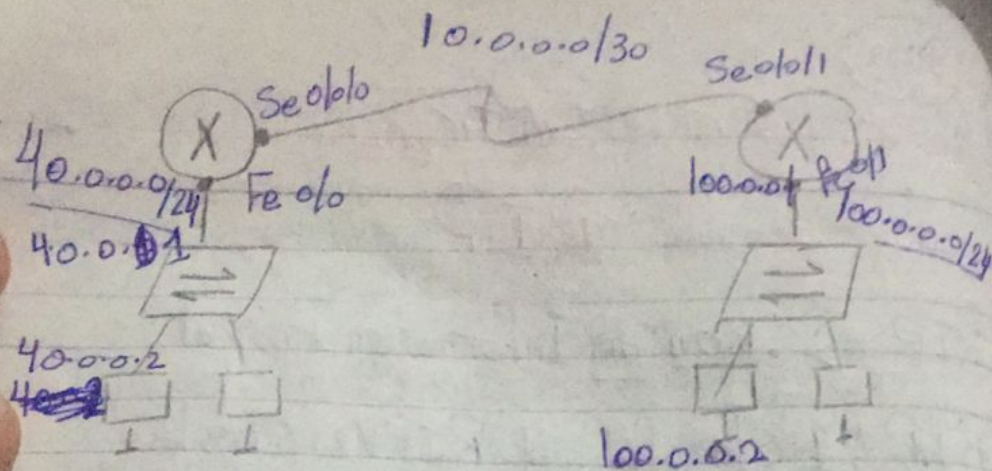
• RIP V1 Classful & RIP V2 Classless

• Administrative Distance (120)

• Period Update (30 sec)

Dead Interval = $6 \times \text{Hello} = 180 \text{ sec}$

• Max HopCount RIP V1 (15)



Cisco Packet Tracker:-

R1 > enable

R1 # conf t

R1(Config) # int Fa0/0

* Fa0/0

R1(Config-int) # ip address 40.0.0.1 255.255.255.0

" # no shutdown

Se0/0/0

R1(Config-int) # ip address 10.0.0.1 255.255.255.0

R1(Config-int) # no shutdown

Se0/0/1

R2(Config-int) # ip address 10.0.0.2 255.255.255.0

R2(Config-int) # no shutdown

0/24
R2(Config-int) # ip address 100.0.0.1 255.255.255.255
Fe 0/1

RIPv2 خطوات

R1(Config) # router rip
" # version 2
R1(Config-rip) # network 40.0.0.0
R1(Config-rip) # network 10.0.0.0
R1(Config-rip) # exit
R1(Config) # do sh ip route
R1(Config) # do write (do copy run start)

إليك مع
شبكة R2 في

eNSP

255.0 [Huawei] int Fe 0/0

هناك الالات عاين م
RIPv2 خطوات

255.2 [Huawei] rip2

[Huawei-rip] network 40.0.0.0

[Huawei-rip] network 10.0.0.0

[Huawei-rip] quit

255.255 [Huawei] display ip routing-table

[Huawei] save

إليك مع شبكة R2 في

EIGRP

EIGRP → Enhanced Interior Gateway Routing Protocol

MultiCast IP? 224.0.0.10

Max Hops? 224

Hello? 5 sec. DeadTime? 3Hello = 15 sec.

Algorithm? Dual Algorithm

Hybrid RP

RIP

OSPF

Remark: RIP MultiCast IP 224.0.0.9

خفوات تلوين Routing Table

[1] Neighbor Discovery

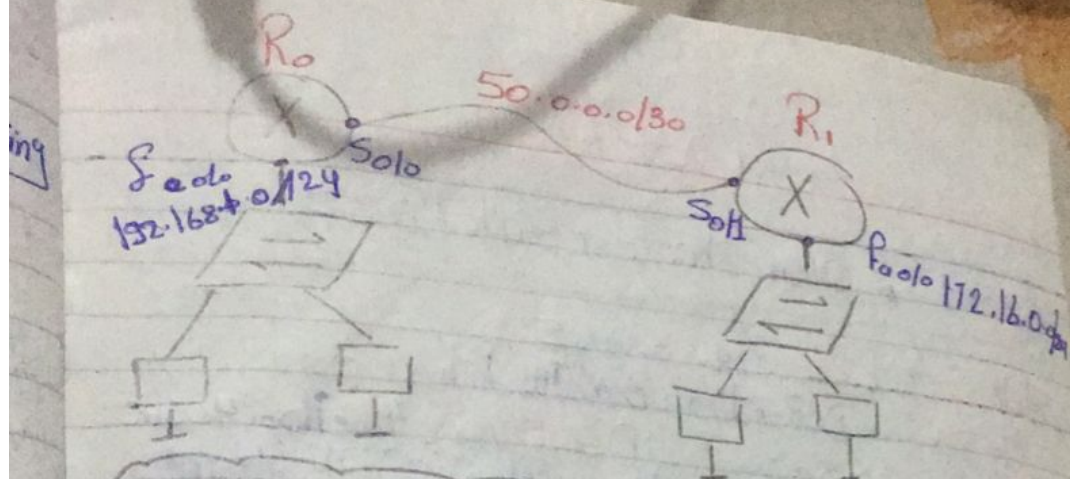
[2] Topology Table

Successor

Possible Successor

[3] Routing Table

Only Successor



Note → Wild Card Mask → $\rightarrow$ hosts عدد
يعني لو عندك $255.0.0.0$
يعني $0.255.255.255$

PKT:

R0(config) # Router eigrp (1)

network 192.168.1.0

0.0.0.255 wildcard mask

network 50.0.0.0

0.0.0.3 wildcard mask

exit

sh ip eigrp neighbor

sh ip eigrp Topology

sh ip route

بعد متعرف الشبكة زي ما سبق
في طابع اسمها
ASN رقم تعريفها
موجود لكل روترات
الشبكة

OSPF

OSPF → Open Shortest Path First

MultiCast IP? 224.0.0.5

Metric? Depend on Bandwidth

Hello? 10sec

DeadTime? 4Hello = 40sec

Algorithm? Dijkstra Algorithm (SPF Algorithm)

1 Neighbor Discovery

2 Routes Discovery

* Exchange of Updates

LSA

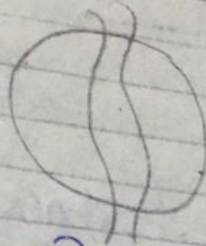
الروتريكي تفاصيل
Configuration

3 LSADB & SPF Tree

يخزنه ال LSADB وليست مترية
LSA

Link	State	Advertisement
9.1.1.1/24	10MBps	W
10.1.1.1/30	64Kbps	
11.1.1.1/30	64Kbps	

LSDB	
9.1.1.1/24	104kBs
10.1.1.1/30	64kBs
11.1.1.1/30	
VI	



Configuration و Area و Process (I) بر فاجتین بر دو

Routing Table

IP	Interface	hops
9.1.1.0/24	E0	0
10.1.1.0/30	S0	0

R → RIP

D → EIGRP

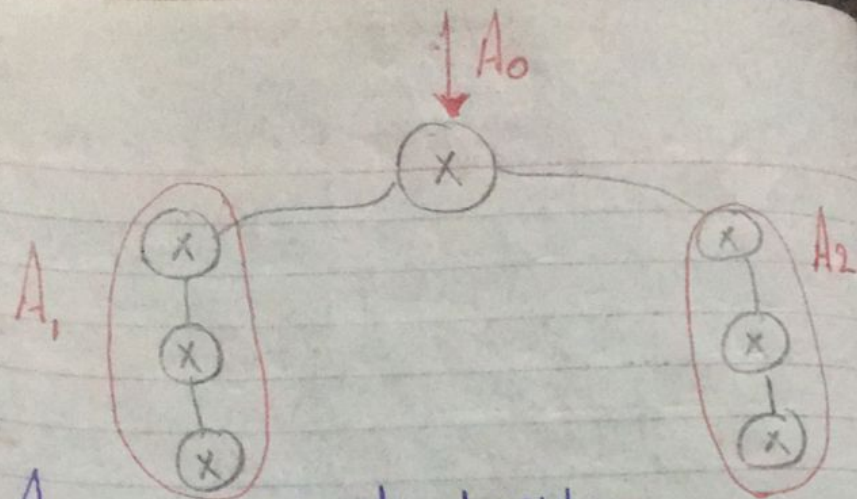
S → Static

O → OSPF

S → Static Routing

C → Connected

مدرعیوبه اند لاکل روتر هیبیت Update بتابعه ل روتر
هیبیت SA بتابعه ده هیبیل Load آیه لکل
لنا بقا هروح اعلی حرکت الحسن اقس Areas



$A_0 \rightarrow$ backbone ستان
 A_0 لو A_1 عازمة Connect ب A_2 لازم قیفت
 بیت اقل استوارک head

Configuration :-
 PKT

(Config) # Router OSPF 1-65535 *Process ID
 # network الشبک wildcard area
 mask

do sh ip route

do sh ip ospf neighbor
 # do sh ip ospf data base

eNSP:-

[H3comer] # OSPF 1

area 0

network

display ip routing table

display ospf

لوروتر متفعل عليه OSPF ، eigrp ، rip هيختار ان

Administrative distance

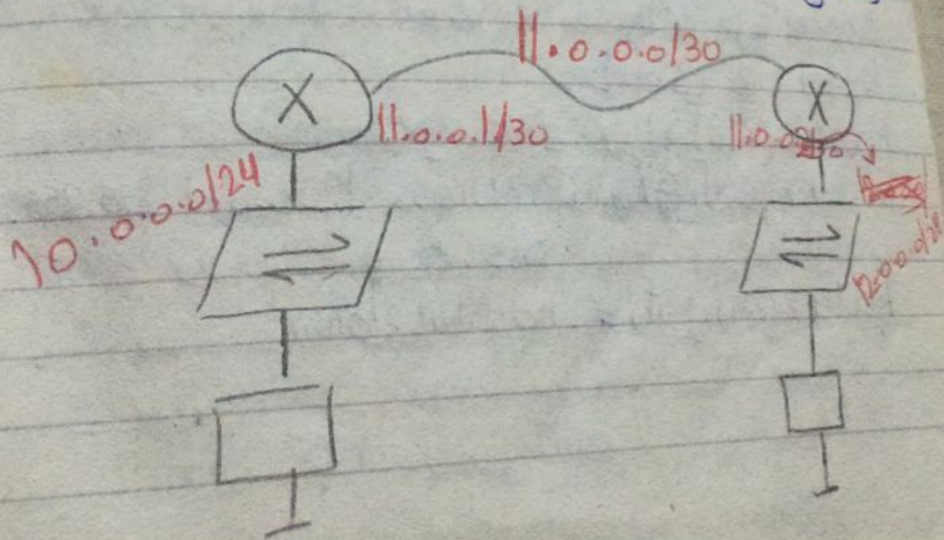
Administrative distance

R → 120

O → 110

D → 90

interior
في الاله انه سفعال Cisco هياف لاقول
لومش Cisco هيشتغل OSPF



DKT

Router> ~~conf t~~ en

Router# conf t

Router(Config) # int g0/0

Router(Config-if) # ip address 10.0.0.1 255.255.255.0

Router(Config-if) # no shutdown

Router(Config) # int s0/3/0

Router(Config-if) # ip address 11.0.0.1 255.255.255.252

Router(Config-if) # no shutdown

end Configuration of Router1

Router2> en

Router2# conf t

Router2(Config) # int g0/0

Router2(Config-if) # ip address 12.0.0.1 255.255.255.0

Router2(Config-if) # no shutdown

Router2(Config)# int se 0/3/0
Router2(Config-if)# ip address 11.0.0.2 255.255.255.0
Router2(Config-if)# no shutdown

Router2(Config-if)# do sh ip route

~~Router2(Config)#~~ # OSPF

Router2(Config)# OSPF 1

Router2(Config-router)# network 12.0.0.0 0.0.0.255
area 0

55. Router2(Config-router)# network 11.0.0.0 0.0.0.255
area 0

Router (Config)# OSPF 1

ter1 Router (Config-router)# network 10.0.0.0 0.0.0.255
area 0

Router (Config-router)# network 11.0.0.0 0.0.0.255
area 0

Router (Config)# sh ip route

5.255. Router (Config)# do sh ip OSPF neighbor

Router (Config)# do sh ip OSPF database

Router1

Configuration # ENSP #

Utl Remove
Utl mlog

[Huawei] int eth 0/0/0
[Huawei] ip address 10.0.0.1 24
[Huawei] int eth 0/0/1
[Huawei] ip address 11.0.0.1 30

Router1

Configuration Router2

[Huawei] int eth 0/0/0
[Huawei] ip address 11.0.0.2 30
[Huawei] int eth 0/0/1
[Huawei] ip address 12.0.0.1 24
[Huawei] display ip-routing table

Router2

OSPF Configuration

[Huawei] ospf 1
[Huawei] area 0
[Huawei] network 12.0.0.0 0.0.0.255
[Huawei] network 11.0.0.0 0.0.0.255

Router1 Cost

Router2

[Huawei] ospf 1

[Huawei] area 0

[Huawei] network 11.0.0.0 0.0.0.3

[Huawei] network 10.0.0.0 0.0.0.255

[Huawei] display ip routing-table

[Huawei] display ospf peer

[Huawei] display ospf lsdb

ospf Router2

WAN (Multi Pl Areas)

WAN

Private

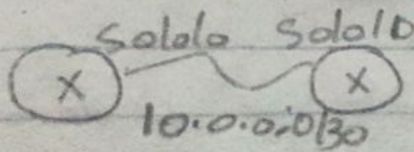
Public

DSL Catv
VSAT

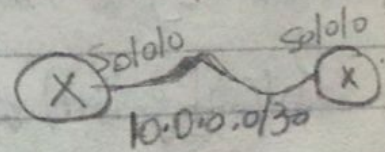
Circuit
↳ HDLC
PPP
↳ ISO
Cisco

Packet

in PKT:-



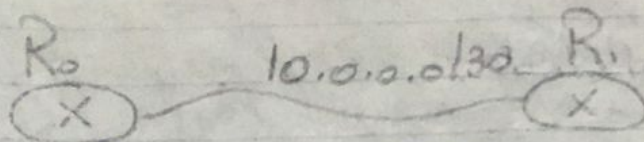
in ENSP:-



Ro(Config-Int) # encapsulation	[Huawei-int]
Hdlc or PPP	link-Protocol Hdlc

الخطوات الجاية على شأن اوعاين العمل
Telnet من جهاز تابع قام كده ركز بقا معايا

in PRT:



R0(Config) # line vty 0-4 * 11 of users.

R0(Config-line) # Password Cisco

R0(Config-line) # login local

R0(Config-line) # exit

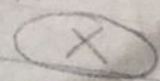
R0(Config) # Service Password-Encryption

R0(Config) # enable ~~secret~~ secret Cisco

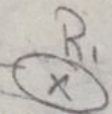
تشفير بجوارزمية 7

تشفير بـ MD5 Hash

in OSPF: - R0



10.0.0.0/30

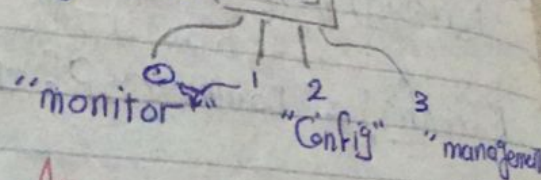
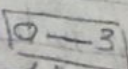


[Huawei] user-interface vty 0-4

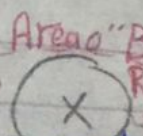
[Huawei] authentication-mode Password

[Huawei] set authentication Password Cipher Cisco

[Huawei] User Privilege level



OSPF Multiple Area: - PKT:



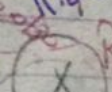
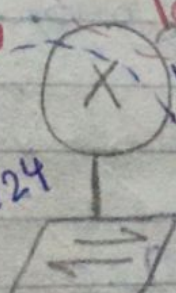
Area "Backbone"



10.0.0.0/30
10.0.0.1/30

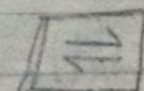
Area 1

192.168.1.0/24



Area 2

192.168.2.0/24



PKT:-

Router 1

Router1 > enable

Router1 # Conf t

Router1 (Config) # int Fa 0/0

Router1 (Config-int) # ip address 192.168.1.1
255.255.255.0

Router1 (Config-int) # no Shut down

Router1 (Config) # int S0/0/0

Router1 (Config-int) # ip address 10.0.0.1
255.255.255.252

Router1 (Config-int) # no Shut down

Router1 (Config) # router ospf 1

Router1 (Config-router) # router-id 1.1.1.1

(virtual int) loopback

Router1 (Config-router) # network 192.168.1.0
0.0.0.255 area 1

Router1 (Config-router) # network 10.0.0.0 0.0.0.3
area 0

#Router 2

Router2 > enable

Router2 # Conf t

Router2 (Config) # int S0/0/0

Router2 (Config-int) # ip address 10.0.0.1 255.255.
255.252

Router2 (Config-int) # no shutdown

Router2 (Config) # int S0/0/1

Router2 (Config-int) # ip address 11.0.0.1 255.255.255.0

Router2 (Config-int) # no shutdown

Router2 (Config) # int S0/0/2

Router2 (Config-int) # ip address 11.0.0.2 255.255.255.
.252

Router2 (Config-int) # ~~no~~ no shutdown

OSPF

Router2 (Config) # router ospf 1

Router2 (Config-router) # router-id 2.2.2.2

Router2 (Config-router) # network 10.0.0.0

25.0.0.3 area 0

Router2 (Config-router) # network 11.0.0.0

0.0.0.3 area 0

#Router3

```
Router3 > enable
Router3 # conf t
Router3(Config) # int Fa0/0
Router3(Config-int) # ip address 192.168.2.1
255.255.255.0
Router3(Config) # int Se0/0/0
Router3(Config-int) # ip address 11.0.0.2 255.255.
.255.252
Router3(Config-int) # no shutdown
```

OSPF

```
Router3(Config) # router ospf 1
Router3(Config-router) # router-id 3.3.3.3
Router3(Config-router) # network 192.168.2.0
0.0.0.255 area 2
Router3(Config-router) # network 11.0.0.0
0.0.0.3 area 0
```

on each Router

```
Router (Config) # show ip ospf neighbor
Router # show ip route
Router # show ip ospf database
```

#telnet

Router(Config) # username admin Password
Cisco

Router(Config) # line Vty 0 4

Router(Config-line) # login local

Router(Config-line) # transport input telnet

Router(Config) # service Password-encryption

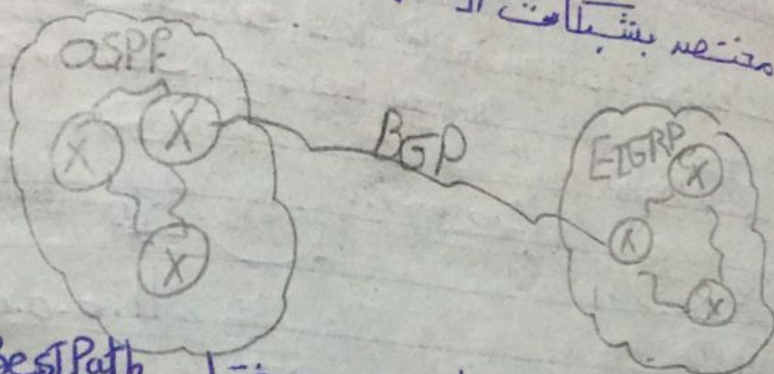
~~Router~~ Router(Config) # enable secret Cisco

BGP

BGP → Border Gateway Protocol

هو المفروض في TCP/IP، والله انتشر عليه قويه

• EXTERIOR → WAN معتصم بشبكات الـ



Best Path BGP مش بيختار اقص طريق - بيختار
بناء على Policies انواع

BGP

EBGP

"EXTERIOR"

Administrative

Distance? 200

AD كل ما قل

IBGP

"Interior"

Administrative

Distance? 200

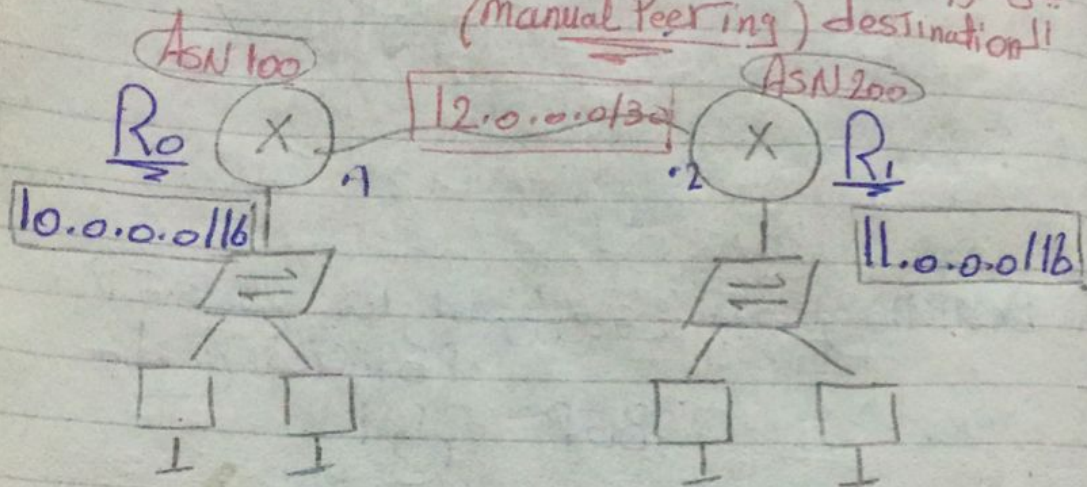
تقل كلما قل
Priority

1] Keep Alive message = 60 second

2] Hold Time Message = 3 keep Alive = 180 second

Bandwidth Update بدون مميزة جيلة في اعلان او في

Note: BGP is UniCast Protocol يعني لازم انت قادر ان
(manual Peering) destination ال



Cisco Packet Tracker:

R0(Config) # router BGP 100

R0(Config-router) # neighbor 12.0.0.2 remote-as 200

R0(Config-router) # ~~network~~ 10.0.0.0 mask 255.255.0.0

R0(Config-router) # exit

R0(Config) # do sh ip route

R0(Config) # do sh ip bgp

R1(Config) # router bgp 200

R1(Config-router) # neighbor 12.0.0.1 remote-as 100

R1(Config-router) # network 11.0.0.0 mask 255.255.0.0

R1(Config-router) # exit

R1(Config) # do show ip route

R1(Config) # do show ip bgp

cNSP:-

[Huawei] bgp 100

[Huawei-router] Peer 12.0.0.2 as-number 200

[Huawei-router] ~~Peer~~ network 10.0.0.0 ~~16~~

[Huawei-router] 9

[Huawei] display ip-routing table

[Huawei-router] Peer 12.0.0.1 ~~as-number~~ 100

[Huawei-router] network 10.0.0.0 ~~16~~

[Huawei-router] 9

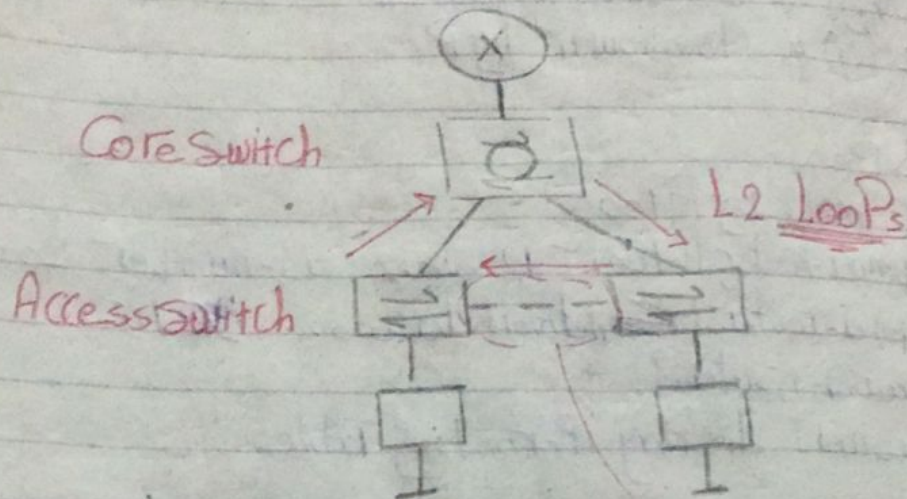
[Huawei] display ip-routing table

as 200

255.0.0

VLAN & STP

STP → Spanning Tree Protocol 21 Dijkstra
(IEEE 802.1d) [Plug & Play]
Dijkstra → EIGRP (D)



Redundancy Backup و التكرار بين الأجهزة
معاً L2 (Loops) يجب عازل الغير العمل ايها
لنا يمنع دور STP

- الروتر مش بيعمل Broadcast بيحمله discard أما
Switch بيعمل يعملي
L2 Loops → Causes Broadcast Storm Problem
← تؤثر على Users

STP? "Spanning Tree Protocol"
STPs?

Convert mesh
ToPo into
StartToPo

علشان جود
Rootswitch

RootPort 11 on non-root switch

Root Port? ! مسخره
 Non-root Switch
 Non-root Port الى Non-root Switch
 بيقول يا Root Port انت بيقول يا Root Port

Root Switch

يبقى الثاني هل هنتار

باقی Ports الی مختار نہاں تبصر فی alternative disable

BPdu → Bridge Protocol Data Unit

Operations:-

1. at Start UP, Hello Message every 2 sec. every Switch send BPdu

Switch ID	Accumulative Path Cost	Port-ID
-----------	------------------------	---------

Select Root Switch → less Switch ID

By default every switch have same Priority 0 — 65535

Switch ID
mac و mac Processor
البيغور و الـ

2. Select Root Port:-

The best Port to unroot switch then can reach root switch based on CSP

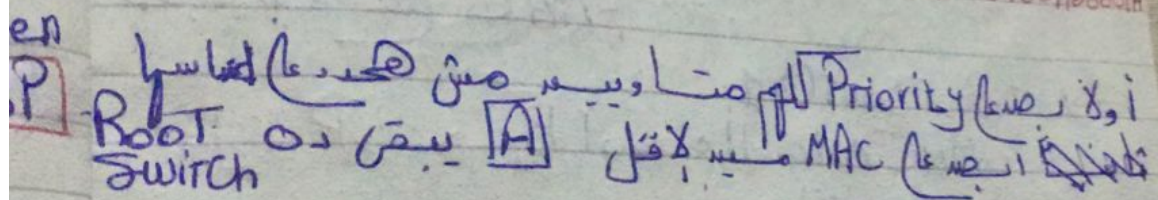
Less Cost ← CSP → Port ID

Accumulative Cost

100m	100m	1G	10G	100G
Cost 100	19	4	2	1

Port ID Root ID
Switch ID
من الـ
الـ
id

3 Select DP.
 2 Ports التناوب فيها
 Root Port الـ DP واحد فيهم
 disable mode هتقفلها وتبقى
 C SP
 Priority 32678
 MAC 000000000000
 Non-Root switch



✓✓
XX

Non-Root Sult Ch[2] $\rightarrow$ $\text{Cost}_{15} = 4 \text{ Cost}_{\text{TV}}$ ✓
 $\text{Cost}_{1+1} = 8 \text{ Cost}_{\text{TX}}$

@Change:-

14:35

at Change

Cable is Failed
Send STP TCN to root
Switch, so only Blocked Port
Entered in, specify it's state
listen

Recovery only after 30 second
لا س لا س بي قبل 15sec في البداية
في الة 15sec + listen في الة
(forced)

Root Failed
Root started to
Send BPDU to
non-root & they
wait for dead
Time 10 hello
 $10 \times 20 = 200 \text{ sec}$
Then short operation
From Begin

Recovery once
after:-
 $2 \times 10 + 30 = 50 \text{ sec}$

فالشئ أنت اختارت ال RP طيب الباقي بقا هييرتقوا
 يسبقوا DP، ولا هتخليهم لعل isabla لعل ال DP
 ال Switchid، Cost، Portid ال هو برودو CSP

Root Switch هتترقي 2 Ports ال Cost=0 لذللك بول
 ال DP (ال Root Switch) دياً هييقن DP ال بورت
 (ساعتها)

ينجى ال Non-Root ←

Non-Root Switch 1;

$$(1G + 1G) \times 8 = \text{Cost}$$

Non-Root Switch 2;

$$(1G + 1G) \times 8 = \text{Cost}$$

طيب بنص على MAC ;

Non-Root Switch 1;

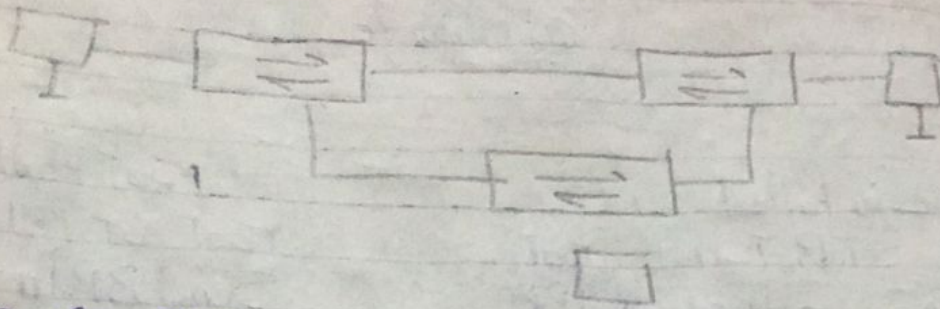
MAC 010011000000 B → DP

Non-Root Switch 2;

MAC 010011000000 C

ال Blocked DP تقع وقتها ما Port

PRT:-



Sw(Config) # do show SPanning-Tree

Sw(Config) # SPanning-Tree vlan1 Priority 0

root Priority

اقل کم فیہ فیقن Root و میس لا. (اجبار)
لیہ VLAN؟ علشان ہو جای سہ الممنوع assigned بکہ
یعنی الی بنادی علی کل Ports.

eNSP:-

[Huawei] stp mode stp

[Huawei] display stp

Experiment VLAN

VLAN → Virtual Local Area Network

Divide Main LAN into Sub LANs Called VLAN Why?
 يعني انما مشي Domain وليك الشروق مايل افسوس
 LAN IT LAN LAN الة لانة LAN المعيشة etc
 VLAN علىشان انه محتاج Broadcast، ترانس، Domain، وكوم ده
 هي ليها عارفة Broadcast، ترانس، Domain، وكوم ده

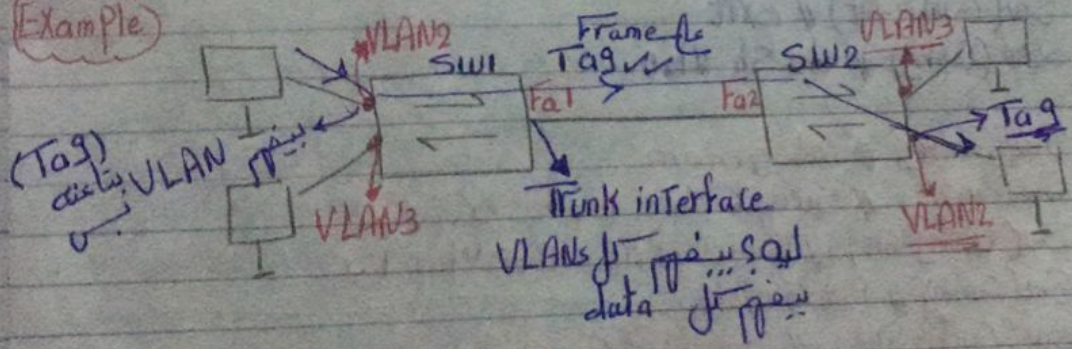
By default Switches و VLAN يكونوا في
 in same Broadcast Domain. يعني لو Device واحد على Broadcast سيجو
 على كل Devices الى على Switch ده.

سؤال يا ليه انا اعمل ترانس بجا Broadcast ده و limited security
 يبقى الخلل VLAN بيشتافليهم على نفس Switch ولله عز وجله
 Virtual ← علىشانهم فتر على نفس Switch.

We Can Divide LAN into Separated VLANs

VLAN Consists of 12 bits => $2^{12} = 4096$ (4096)
 normal use VLAN (100)

Example



eN/SP₁

[Sw] vlan batch 10 20

[Sw] int G0/0/1

[Sw] Port link-type access [Sw] Port default Vlan 10

[Sw] 9

[Sw] int G0/0/2

[Sw] Port link-type access

[Sw] Port ~~link-type~~ default Vlan 20

[Sw] 9

[Sw] int G0/0/0

[Sw] Port link-type trunk

[Sw] Port trunk allow-pass Vlan 10 20

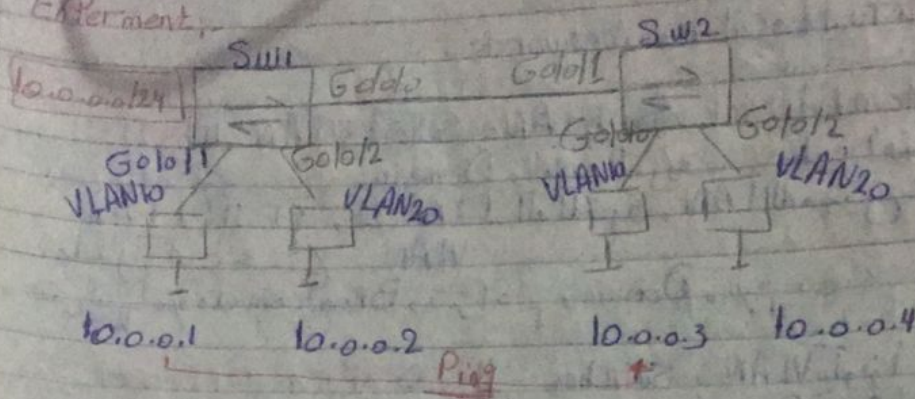
[Sw] 9

[Sw] display Vlan

~~VLAN~~

interVlan Routing

Experiment

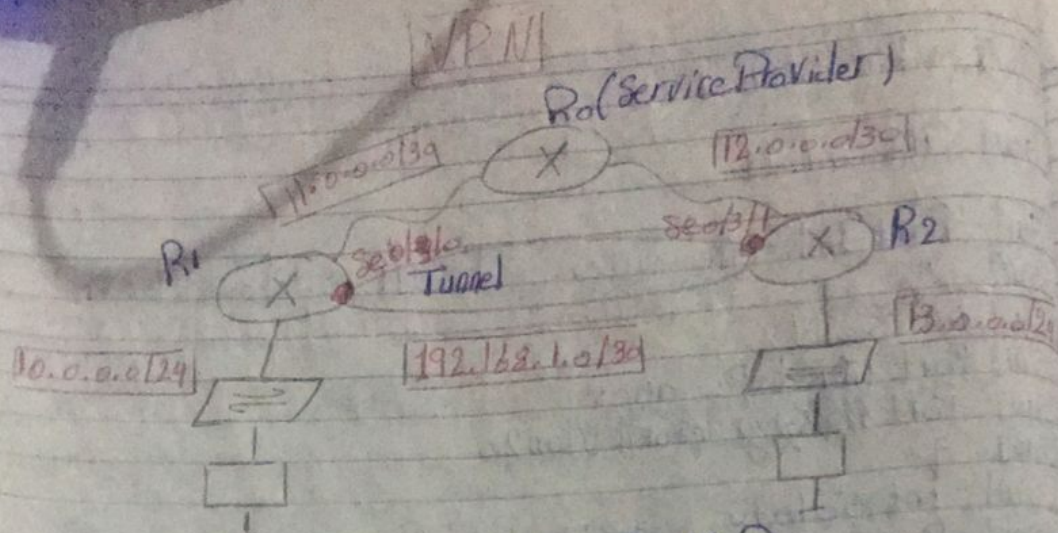


```

Switch 1
Sw1(Config) # int G0/0/1
Sw1(Config-int) # SwitchPort mode access
Sw1(Config-int) # SwitchPort access Vlan 10
Sw1(Config-int) # exit
Sw1(Config) # int G0/0/2
Sw1(Config-int) # SwitchPort mode access
Sw1(Config-int) # SwitchPort access Vlan 20
Sw1(Config-int) # exit
Sw1(Config) # int G0/0/0
Sw1(Config-int) # SwitchPort mode trunk
Sw1(Config-int) # exit
Sw1(Config) # do sh Vlan
    
```

```

Switch 2
Sw2(Config) # int G0/0/0
Sw2(Config) # SwitchPort mode access
Sw2(Config-int) # SwitchPort access Vlan 10
Sw2(Config) # int G0/0/1
Sw2(Config-int) # SwitchPort mode access
Sw2(Config-int) # SwitchPort access Vlan 20
Sw2(Config) # int G0/0/1
Sw2(Config-int) # SwitchPort mode trunk
    
```



After Configuring IPs in PKT & eNSP

PKT:-

R1 Configuration Tunnel

R1(Config) # int tunnel 0

R1(Config-if) # ip address 192.168.1.1 255.255.255.252

R1(Config-if) # tunnel source Se 0/3/0 # int الفاتح

R1(Config-if) # tunnel destination 12.0.0.1 # IP Destination

R2 Tunnel Configuration

R2(Config) # int tunnel 0

R2(Config-if) # ip address 192.168.1.2 255.255.255.252

R2(Config-if) # tunnel source Se 0/3/1

R2(Config-if) # tunnel destination 11.0.0.1 # IP Destination

فيلد جميل الكرم - انا عاوز اعمل حافة - انا عاوز اخلو الى
 الشبكات تروح على Service Provider
 دول بينهم VPN - تقبلوا ازاى ؟ قال IP route
 IP route 0.0.0.0 0.0.0.0 # ip dest ?
 متاع Home Router اهو ده هيفعل في IP اروح على Service Provider.

R1(Config) # IP route

0.0.0.0 0.0.0.0 12.0.0.2

R2(Config) # IP route 0.0.0.0 0.0.0.0 12.0.0.2

Still Tunnel Created li - مشالتي
مش شايف B.0.0.0 و R2 شايف 10.0.0.0
مش شايف R1. مش شايف B.0.0.0
مش شايف R2. مش شايف R1. مش شايف B.0.0.0

R2 - OSPF

R2(Config) # router ospf 1

R2(Config-router) # network 192.168.1.0 0.0.0.3 area 0

R2(Config-router) # network 13.0.0.0 0.0.0.255 area 0

R1 - OSPF

R1(Config) # router ospf 1

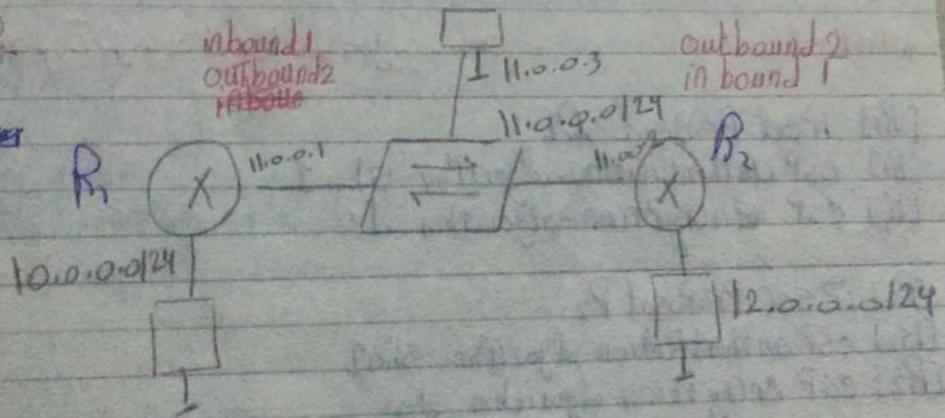
R1(Config-router) # network 10.0.0.0 0.0.0.255 area 0

R1(Config-router) # network 192.168.1.0 0.0.0.3 area 0

do sh IP route

ENSP

Router



R₂
[Huawei] ip route-static 10.0.0.0 24 9.0.0.0 or 11.0.0.1
int R₂ next hop

R₁
[Huawei] ip route-static 12.0.0.0 24 9.0.0.0 or 11.0.0.2
int R₁ next hop

* Creating ACL in R₁ & R₂

R₁
[R₁] acl number 3002
rule 10 permit ip source 10.0.0.0 0.0.0.255
destination 12.0.0.0 0.0.0.255

R₂
[R₂] acl number 3001
rule 5 permit ip source 12.0.0.0 0.0.0.255 destination
10.0.0.0 0.0.0.255

* Proposal adding Header 3DES, sha1

[R₁] ipsec Proposal P1
[R₁] esp authentication-algorithm sha1
[R₁] esp encryption-algorithm des

[R₂] ipsec Proposal P2
[R₂] esp authentication-algorithm sha1
[R₂] esp encryption-algorithm des

* Create Policy

[R₁] ipsec Policy P1 ~~import~~ manual
[R₁] tunnel local 11.0.0.1
[R₁] tunnel remote 11.0.0.2

11.0.0.1
next hop

[R1] security acl 3002

[R1] Proposal P1

[R1] Sa SPI inbound esp 12345

[R1] Sa SPI Outbound esp 54321

[R1] Sa String-key inbound esp simple sha

[R1] Sa String-key outbound esp simple sha

11.0.0.2
next hop

[R2] ipsec Policy P2 2 manual

[R2] ~~tunnel local~~ 11.0.0.2 Security acl 3001

[R2] Proposal P2

[R2] tunnel local

[R2] tunnel Remote

[R2] Sa SPI inbound esp 54321

[R2] Sa SPI outbound esp 12345

[R2] Sa String-key inbound esp simple sha

[R2] Sa String-key outbound esp simple sha

255

s destination

acl

[R1] int g0/0/0

[R1] ipsec Policy P1

[R2] int g0/0/0

[R2] ipsec Policy P2